ASTANA IT UNIVERSITY

Computer networks



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Lecture 6

IPv4 Addressing

Objectives

IPv4 Address Structure	IPv4 Unicast, Broadcast, and Multicast	Subnet an IPv4 Network

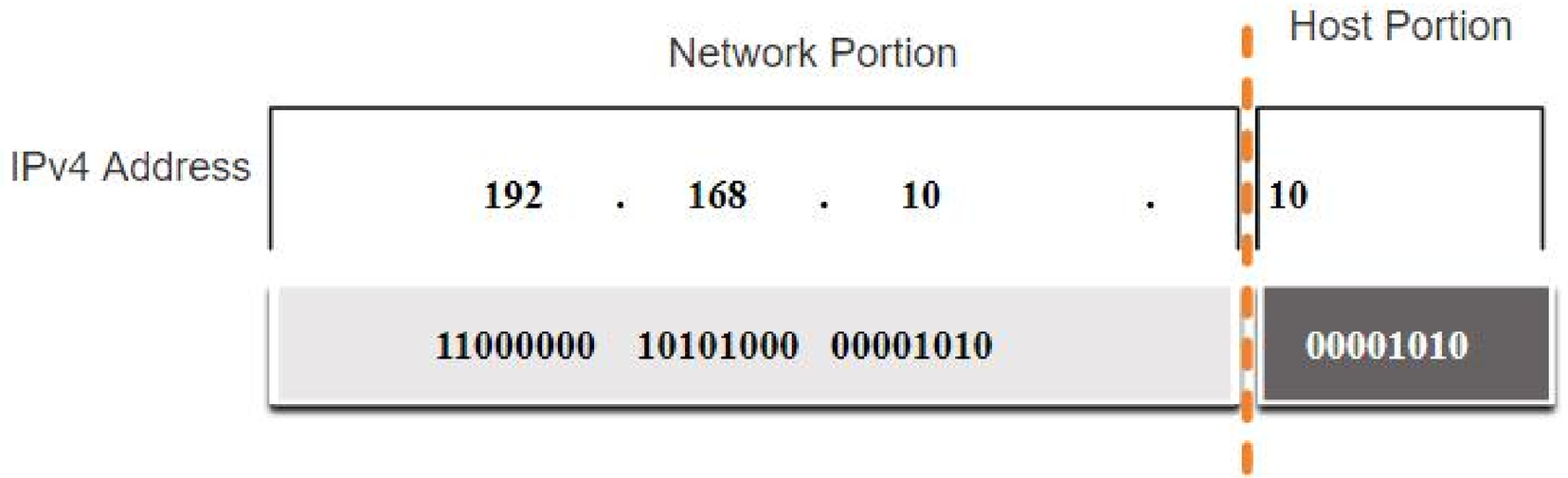
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IPv4 Address Structure

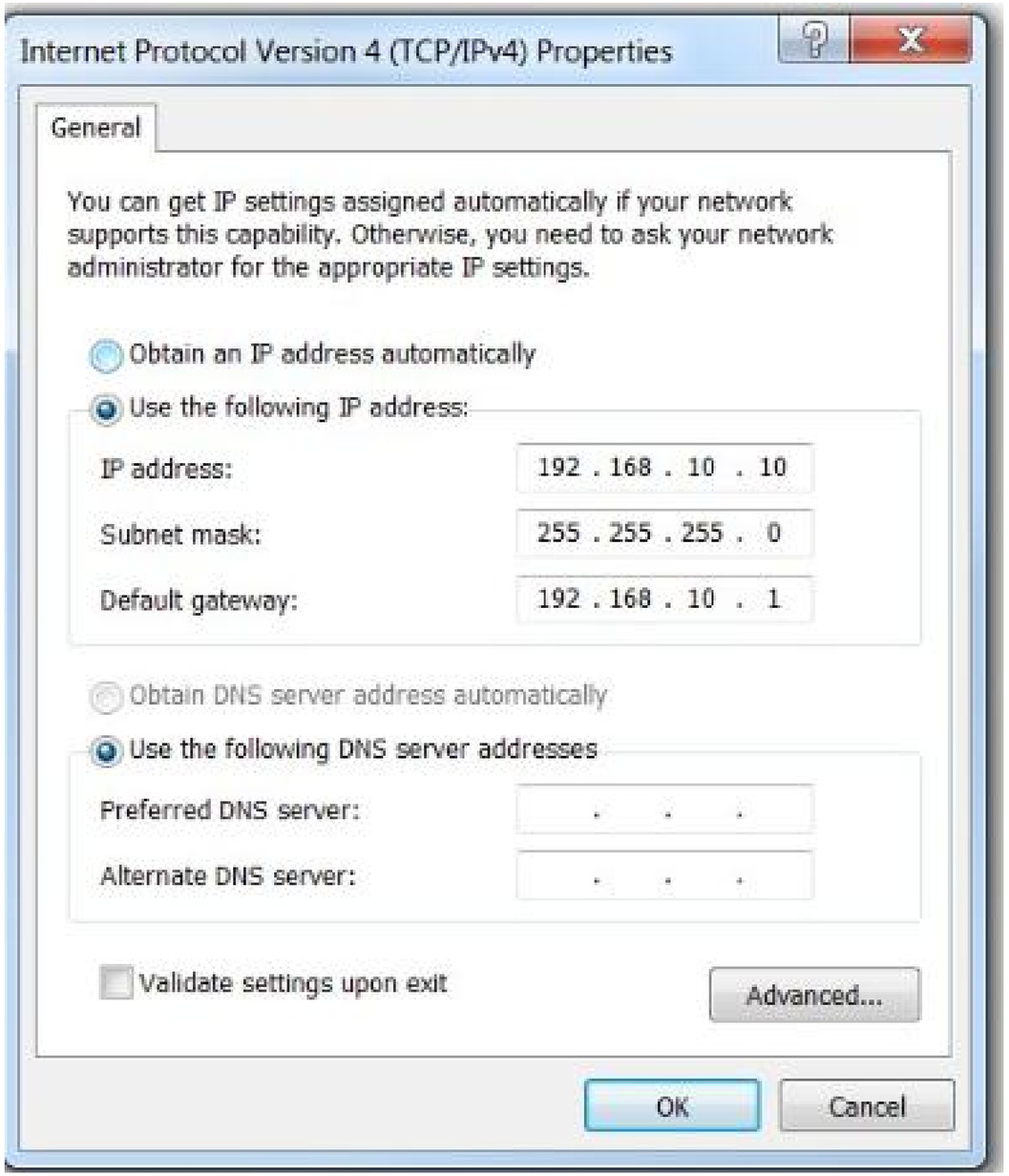
Network and Host Portions

- An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.

- When determining the network portion versus the host portion, you must look at the 32-bit stream.
- A subnet mask is used to determine the network and host portions.



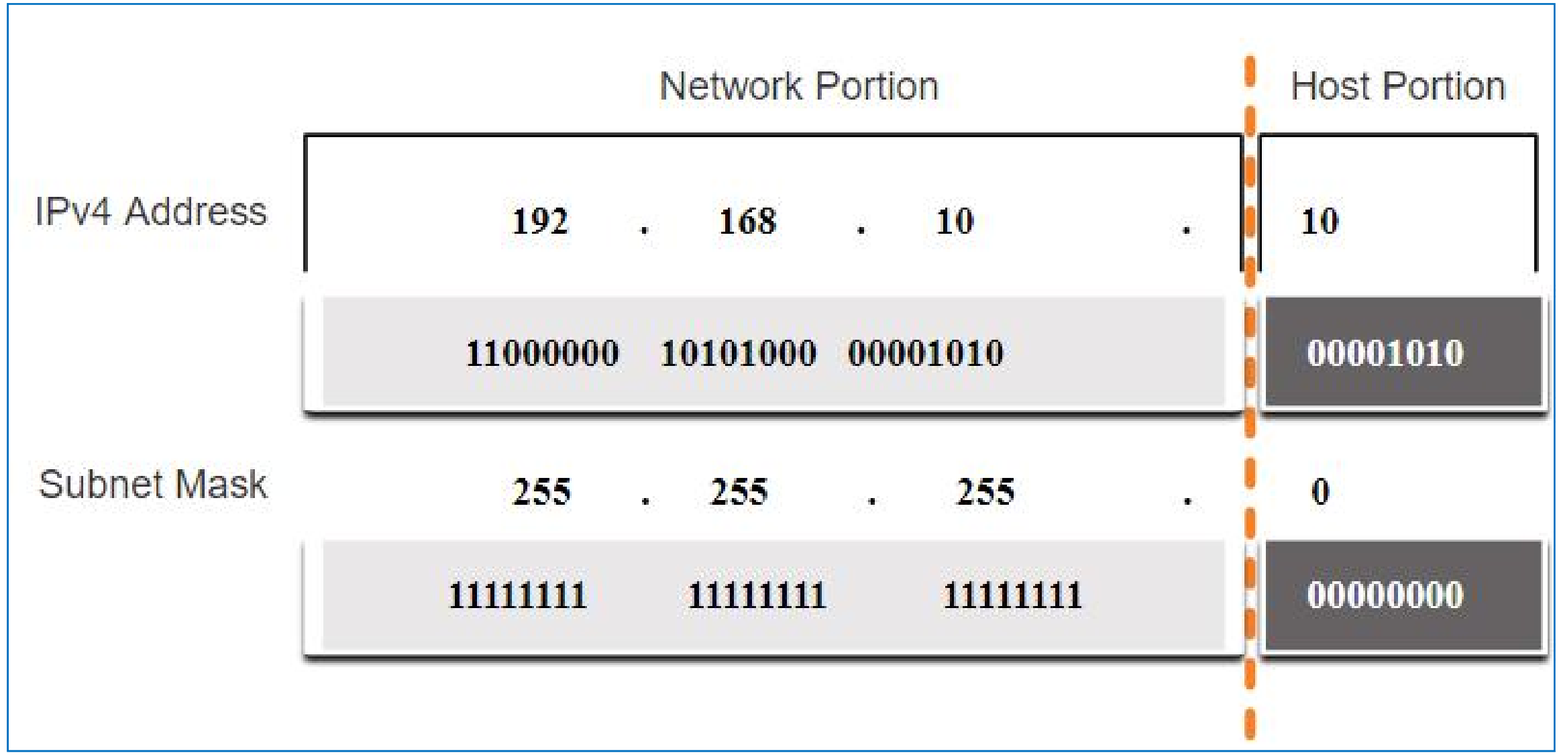
IPv4 Address Structure



The Subnet Mask

- To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.

- The actual process used to identify the network and host portions is called ANDing.



The Prefix Length

- A prefix length is a less cumbersome method used to identify a subnet mask address.

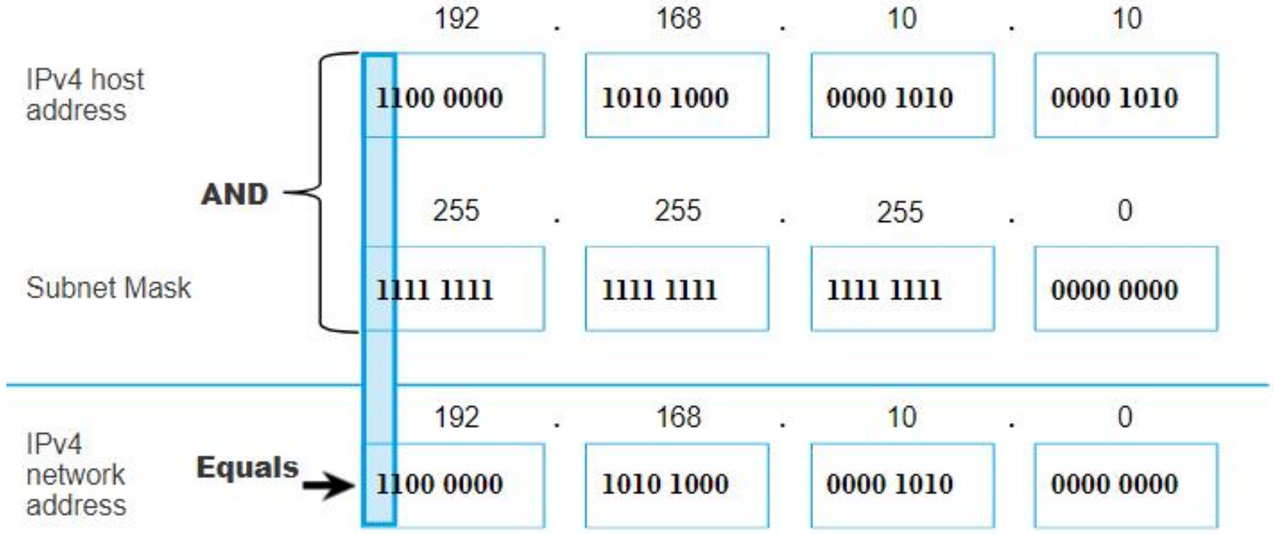
- The prefix length is the number of bits set to 1 in the subnet mask.
- It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

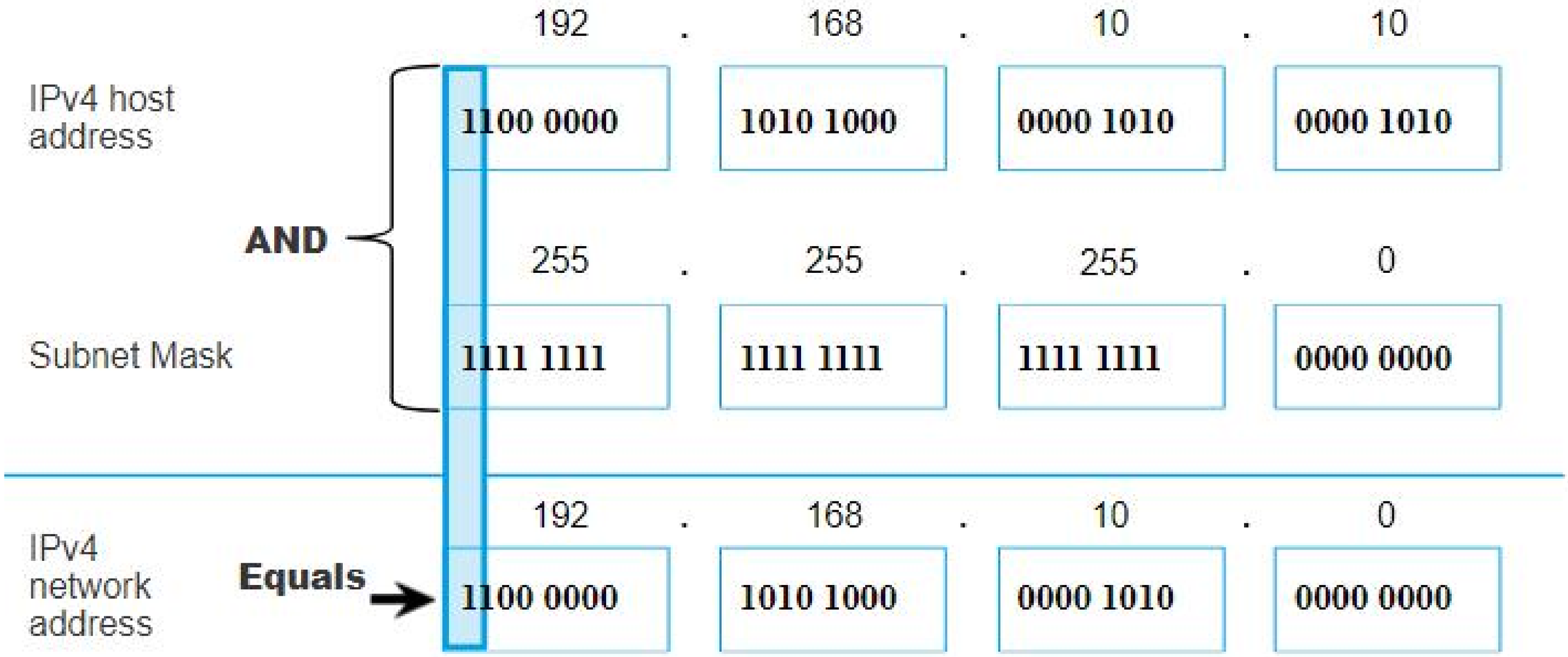
Determining the Network: Logical AND

- A logical AND Boolean operation is used in determining the network address.
  - Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
  - 1 AND 1 = 1, 0 AND 1 = 0, 1 AND 0 = 0, 0 AND 0 = 0
  - 1 = True and 0 = False

- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.



Determining the Network: Logical AND



Important Addresses to Determine

- Network Address (first address in the range)
- Broadcast Address (last address in the range)
- First usable host (address after the network address)
- Last usable host (address before the broadcast address)

Using ANDing to Determine the Network

- A device needs to know what network it belongs to in order to forward data correctly.
- Using their host IP address, their subnet mask, and a process called binary ANDing, a device can find the network that it belongs to.
- To do this, devices compare their host IP and their subnet mask bit-for-bit
  - If the bit values are both a binary 1, the result is a binary 1.
  - If one or both of the bit values is 0, the result is a binary 0.

Determining the Network Address

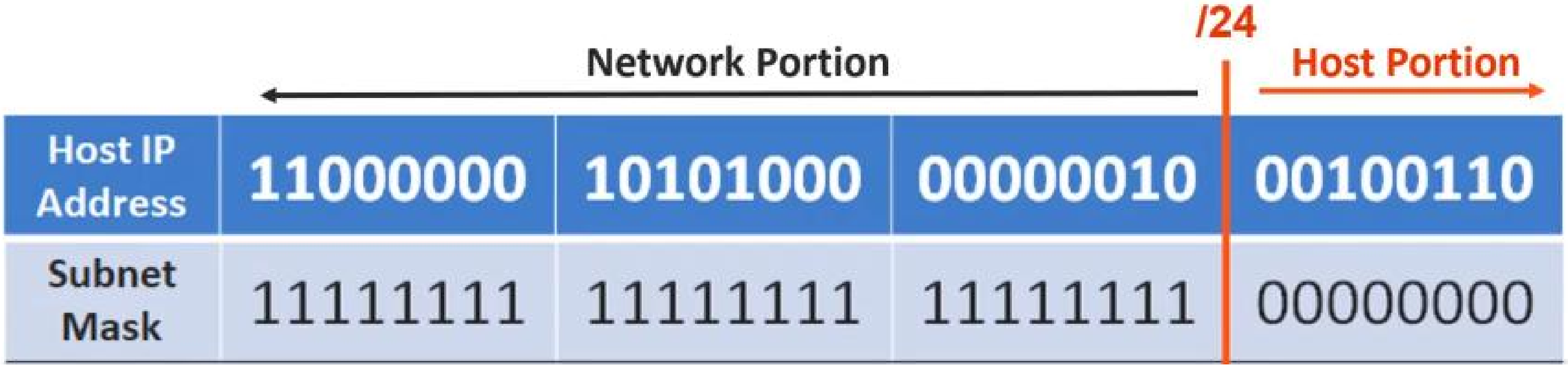
- Given the host IPv4 address and subnet mask 192.168.2.38/24, use ANDing to find the network address of the host.

Determining the Network Address

- Given the host IPv4 address and subnet mask 192.168.2.38/24, use ANDing to find the network address of the host.
  - /24 subnet mask equals 255.255.255.0

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Determining the Network Address

- Given the host IPv4 address and subnet mask 192.168.2.38/24, use ANDing to find the network address of the host.
  - /24 subnet mask equals 255.255.255.0

/24

Network Portion ← → Host Portion

Host IP Address	11000000	10101000	00000010	00100110
Subnet Mask	11111111	11111111	11111111	00000000
Network Address	11000000	10101000	00000010	00000000

Determining the Network Address

- Given the host IPv4 address and subnet mask 192.168.2.38/24, use ANDing to find the network address of the host.
  - /24 subnet mask equals 255.255.255.0

/24

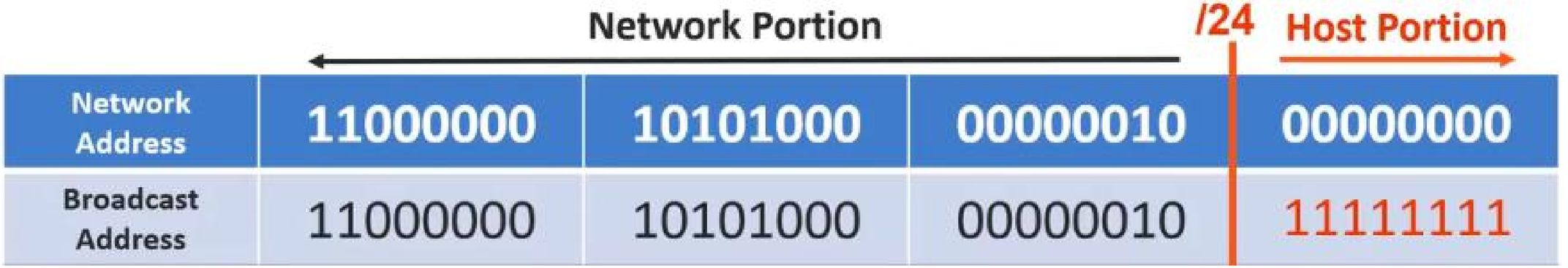
Network Portion ← → Host Portion

Host IP Address	11000000	10101000	00000010	00100110
Subnet Mask	11111111	11111111	11111111	00000000
Network Address	11000000	10101000	00000010	00000000

Network Address = **192.168.2.0/24**

Determining the Broadcast Address

- Used to send a message to all devices on the network at once.
  - Keep the network portion the same
  - Place all binary 1s in the host portion (since the host portion in this example is just the last octet, change all bits in last octet to 1s)
- Convert to dotted-decimal



Network Address	11000000	10101000	00000010	00000000
Broadcast Address	11000000	10101000	00000010	11111111

Determining the Broadcast Address

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Network Portion ← /24 → Host Portion

Network Address	11000000	10101000	00000010	00000000
Broadcast Address	11000000	10101000	00000010	11111111
Broadcast Address Dotted-Decimal	192	168	2	255

- Broadcast Address for this network is **192.168.2.255**

Determining the First Usable Host Address

- Usable host addresses lie between the network address and the broadcast address
- First usable host in binary will be all binary 0s with a binary 1 at the end of the host portion, then convert to dotted-decimal.

Determining the First Usable Host Address

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Network Portion ← /24 → Host Portion

Network Address	11000000	10101000	00000010	00000000
First Usable Host Address	11000000	10101000	00000010	00000001

Determining the First Usable Host Address

- Usable host addresses lie between the network address and the broadcast address
- First usable host in binary will be all binary 0s with a binary 1 at the end of the host portion, then convert to dotted-decimal.

Network Portion ← /24 → Host Portion

Network Address	11000000	10101000	00000010	00000000
First Usable Host Address	11000000	10101000	00000010	00000001
First Usable Host Dotted-Decimal	192	168	2	1

- First usable host for this network is **192.168.2.1**

Determining the Last Usable Host Address

- Last usable host in binary will be all binary 1s with a binary 0 at the end of the host portion, then convert to dotted-decimal.
  - Note: This is the opposite bit pattern of the first usable host

Determining the Last Usable Host Address

- Last usable host in binary will be all binary 1s with a binary 0 at the end of the host portion, then convert to dotted-decimal.
  - Note: This is the opposite bit pattern of the first usable host

Network Portion ← /24 → Host Portion

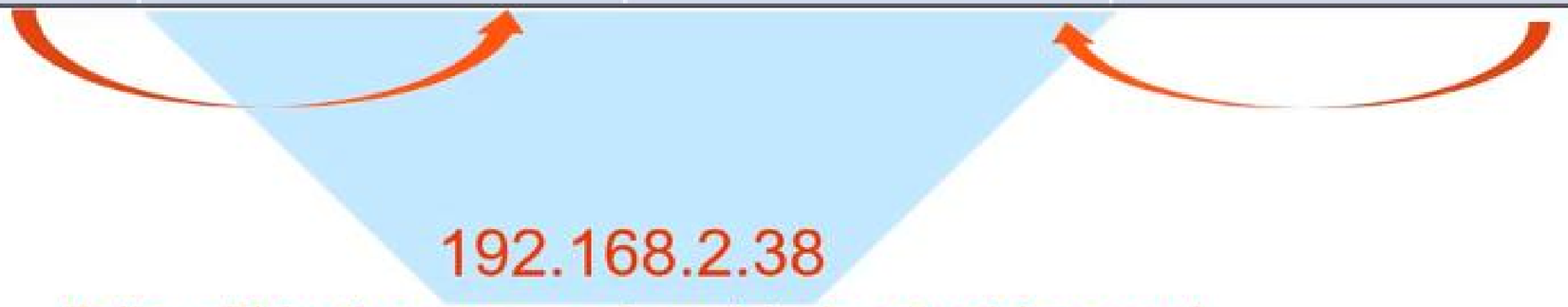
Network Address	11000000	10101000	00000010	00000000
Last Usable Host Address	11000000	10101000	00000010	11111110
Last Usable Host Dotted-Decimal	192	168	2	254

- Last usable host for this network is **192.168.2.254**

To Recap Our Calculations

- Given the host IP address of 192.168.2.38/24, we determined:

Network Address	First Usable Host	Last Usable Host	Broadcast Address
192.168.2.0	192.168.2.1	192.168.2.254	192.168.2.255

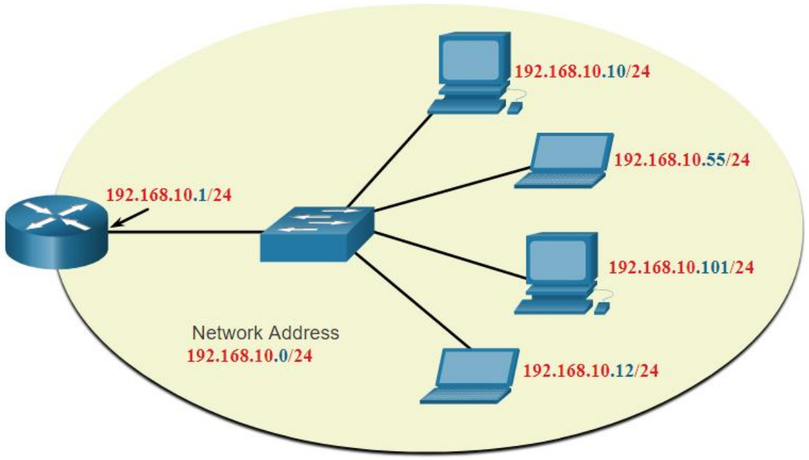


IPv4 Address Structure

Network, Host and Broadcast Addresses

Network, Host, and Broadcast Addresses

- Within each network are three types of IP addresses:
  - Network address
  - Host addresses
  - Broadcast address



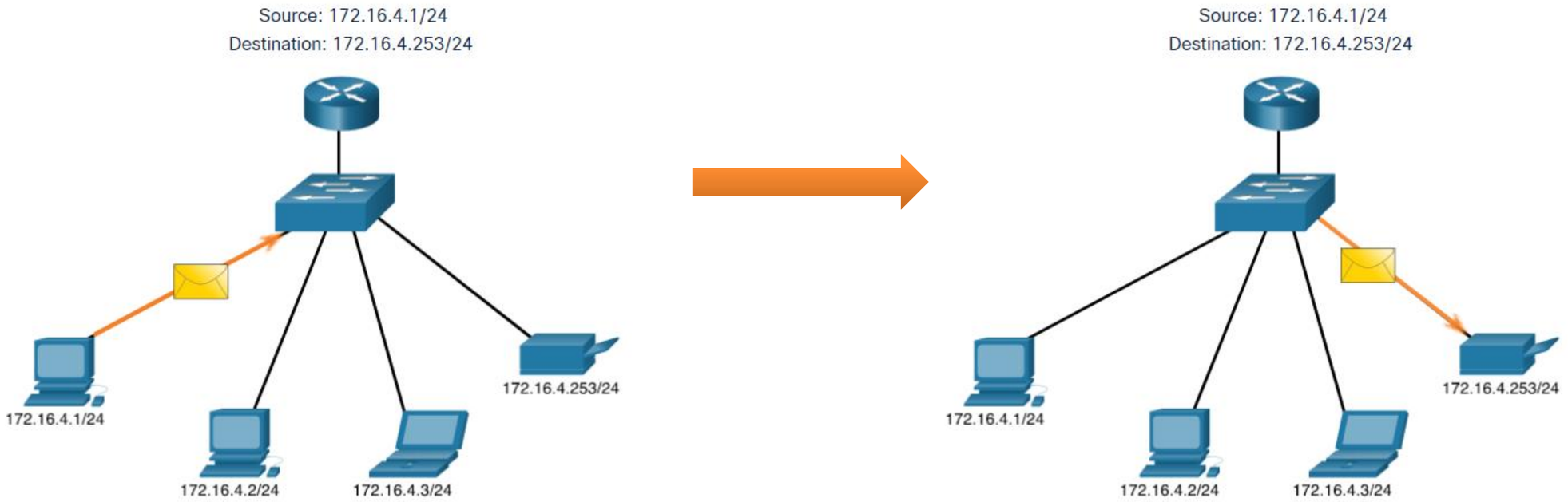
	Network Portion			Host Portion	Host Bits
Subnet mask **255.255.255.**0 or **/24**	255 11111111	255 11111111	255 11111111	0 00000000	
Network address **192.168.10.**0 or **/24**	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
First address **192.168.10.**1 or **/24**	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
Last address **192.168.10.**254 or **/24**	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
Broadcast address **192.168.10.**255 or **/24**	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

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IPv4 Unicast, Broadcast, and Multicast

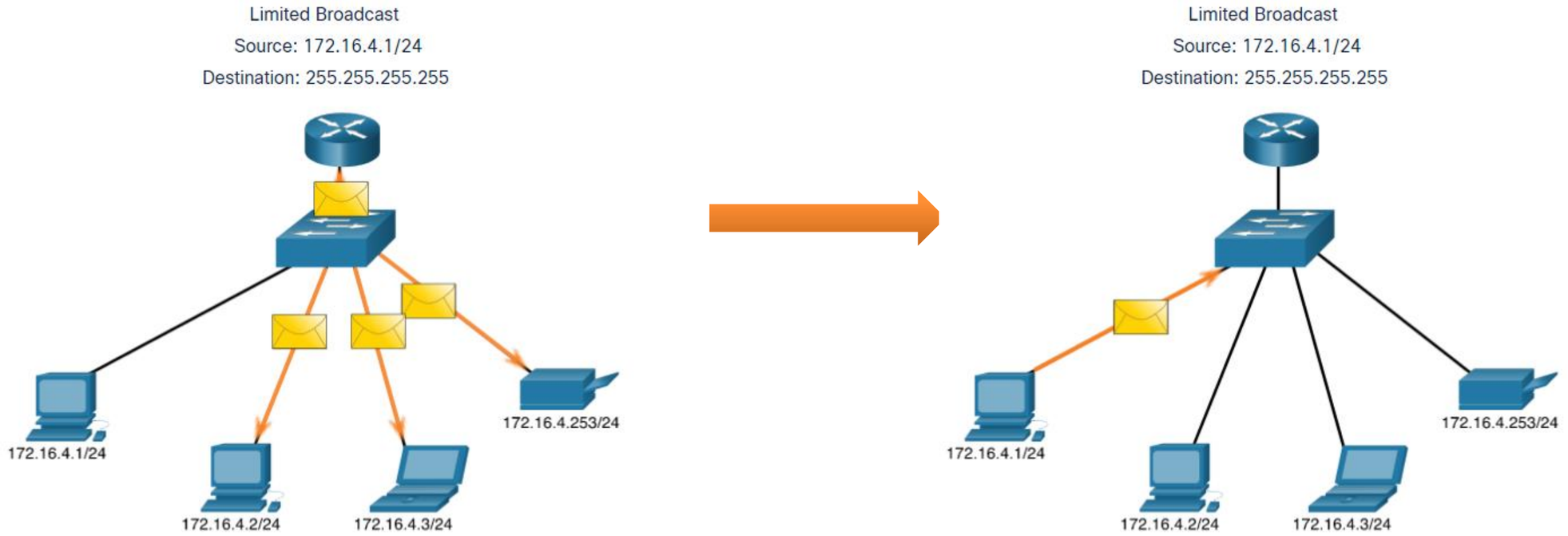
Unicast

- Unicast transmission is sending a packet to one destination IP address.
- For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253.



Broadcast

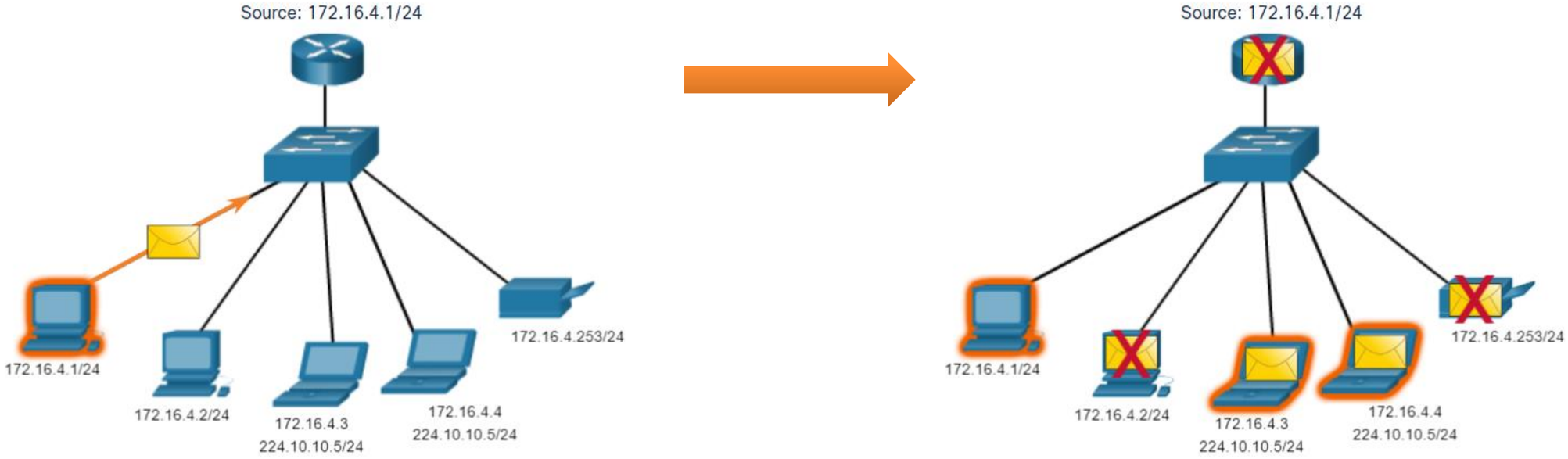
- Broadcast transmission is sending a packet to all other destination IP addresses.
- For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.

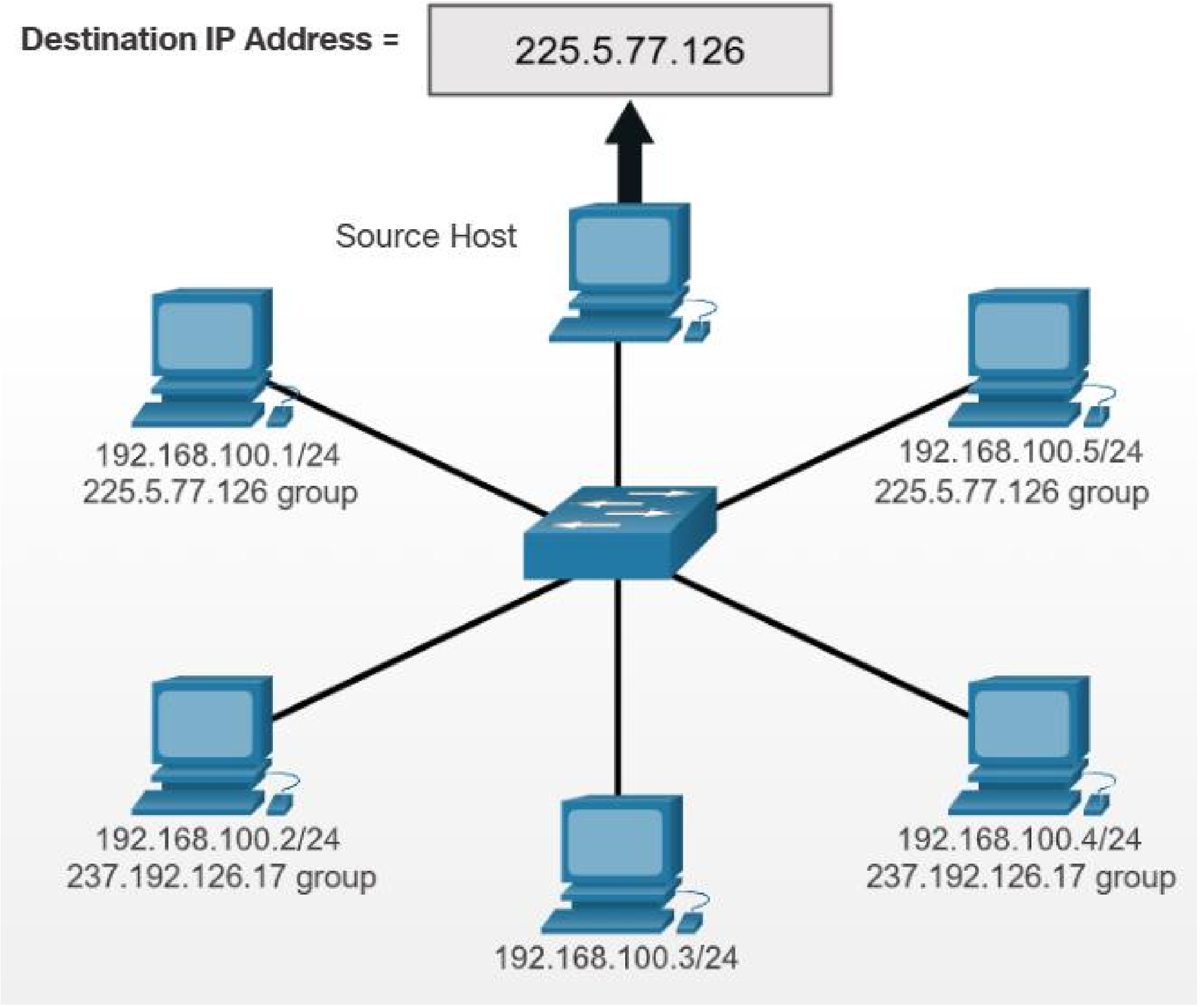


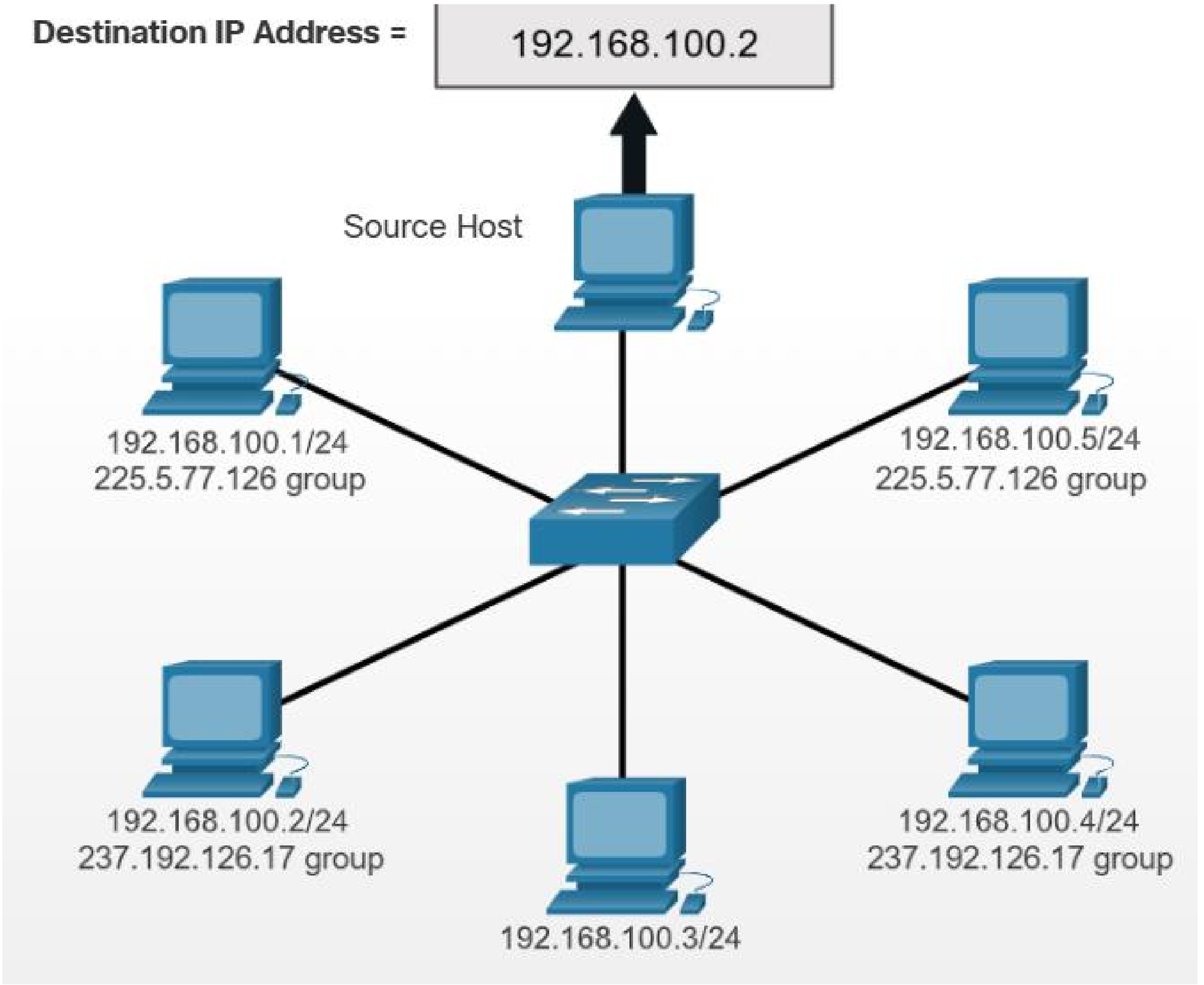
Multicast

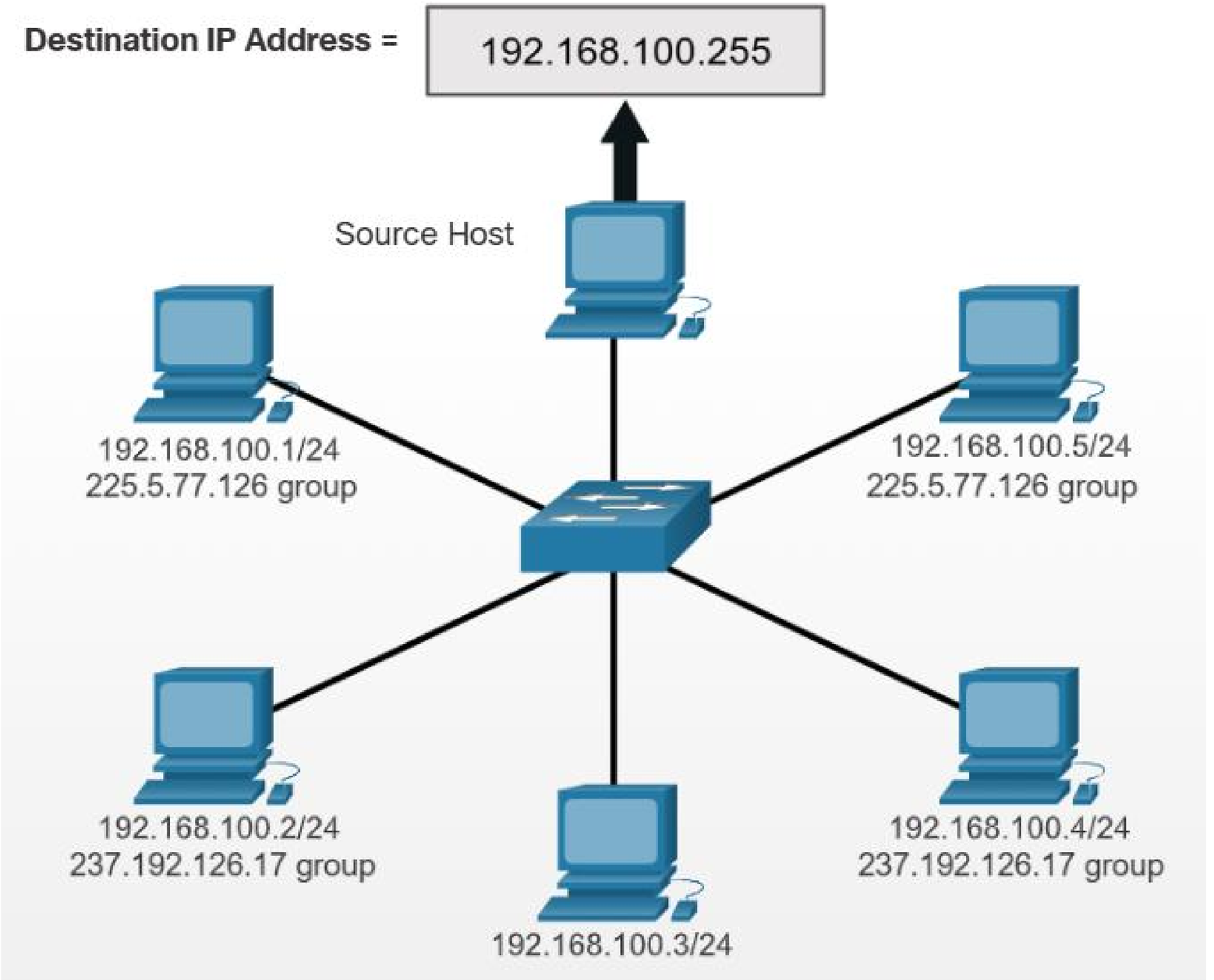
- Multicast transmission is sending a packet to a multicast address group. A multicast packet is a packet with a destination IP address that is a multicast address. IPv4 has reserved the 224.0.0.0 to 239.255.255.255 addresses as a multicast range.

- For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.




Destination IP Address =
225.5.77.126
Source Host
192.168.100.1/24
225.5.77.126 group
192.168.100.5/24
225.5.77.126 group
192.168.100.2/24
237.192.126.17 group
192.168.100.3/24
192.168.100.4/24
237.192.126.17 group


Destination IP Address =
192.168.100.2
Source Host
192.168.100.1/24
225.5.77.126 group
192.168.100.5/24
225.5.77.126 group
192.168.100.2/24
237.192.126.17 group
192.168.100.3/24
192.168.100.4/24
237.192.126.17 group


Destination IP Address =
192.168.100.255
Source Host
192.168.100.1/24
225.5.77.126 group
192.168.100.5/24
225.5.77.126 group
192.168.100.2/24
237.192.126.17 group
192.168.100.3/24
192.168.100.4/24
237.192.126.17 group

Types of IPv4 Addresses

Public and Private IPv4 Addresses

- As defined in in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.

- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.

- Private IPv4 addresses are not unique and can be used internally within any network.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

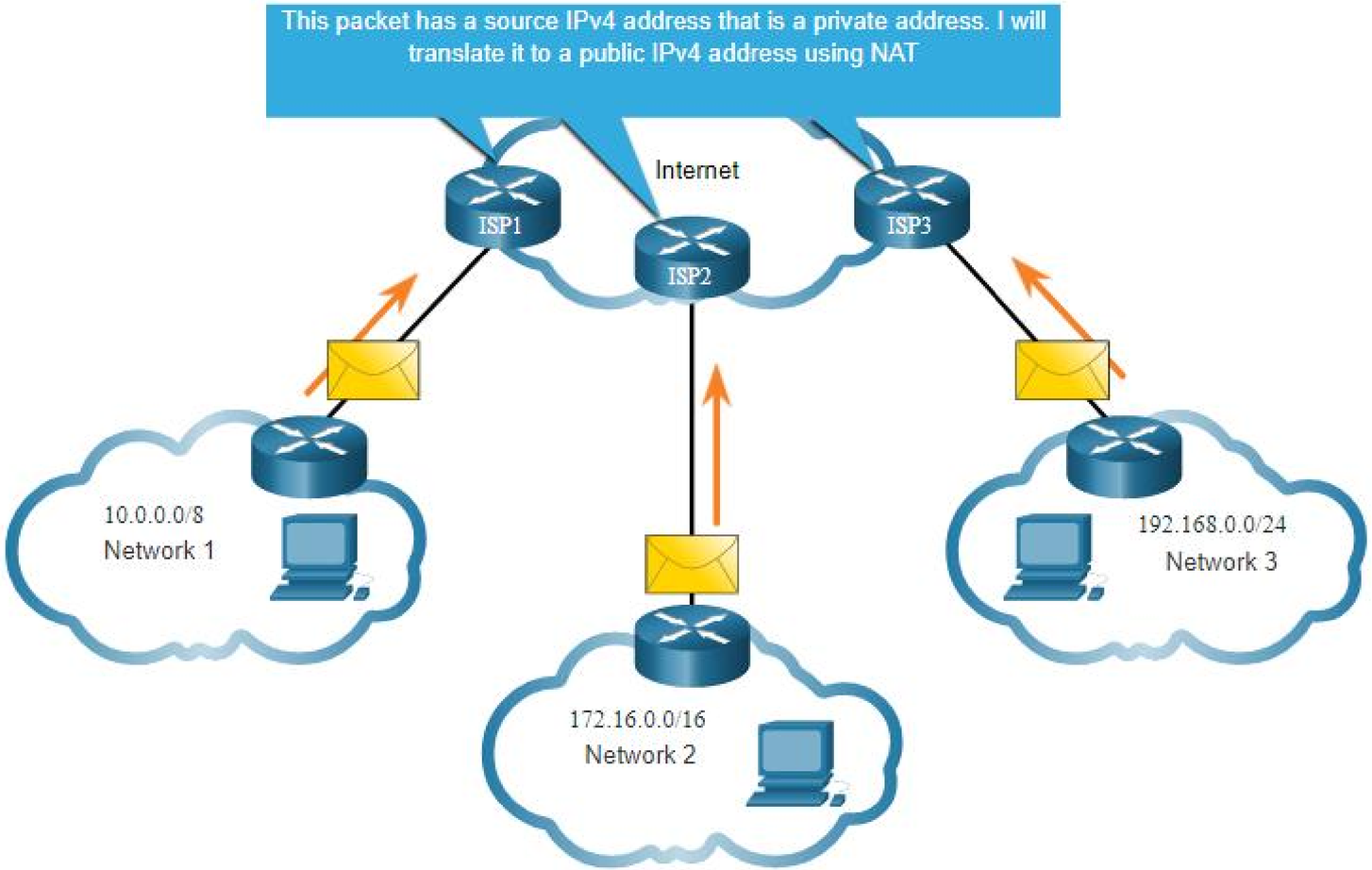
- However, private addresses are not globally routable.

Routing to the Internet

- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.

- NAT is typically enabled on the edge router connecting to the internet.

- It translates the internal private address to a public global IP address.



Special Use IPv4 Addresses

Loopback addresses

- 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
- Commonly identified as only 127.0.0.1
- Used on a host to test if TCP/IP is operational.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

Link-Local addresses

- 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
- Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- Used by Windows DHCP clients to self-configure when no DHCP servers are available.

Legacy Classful Addressing

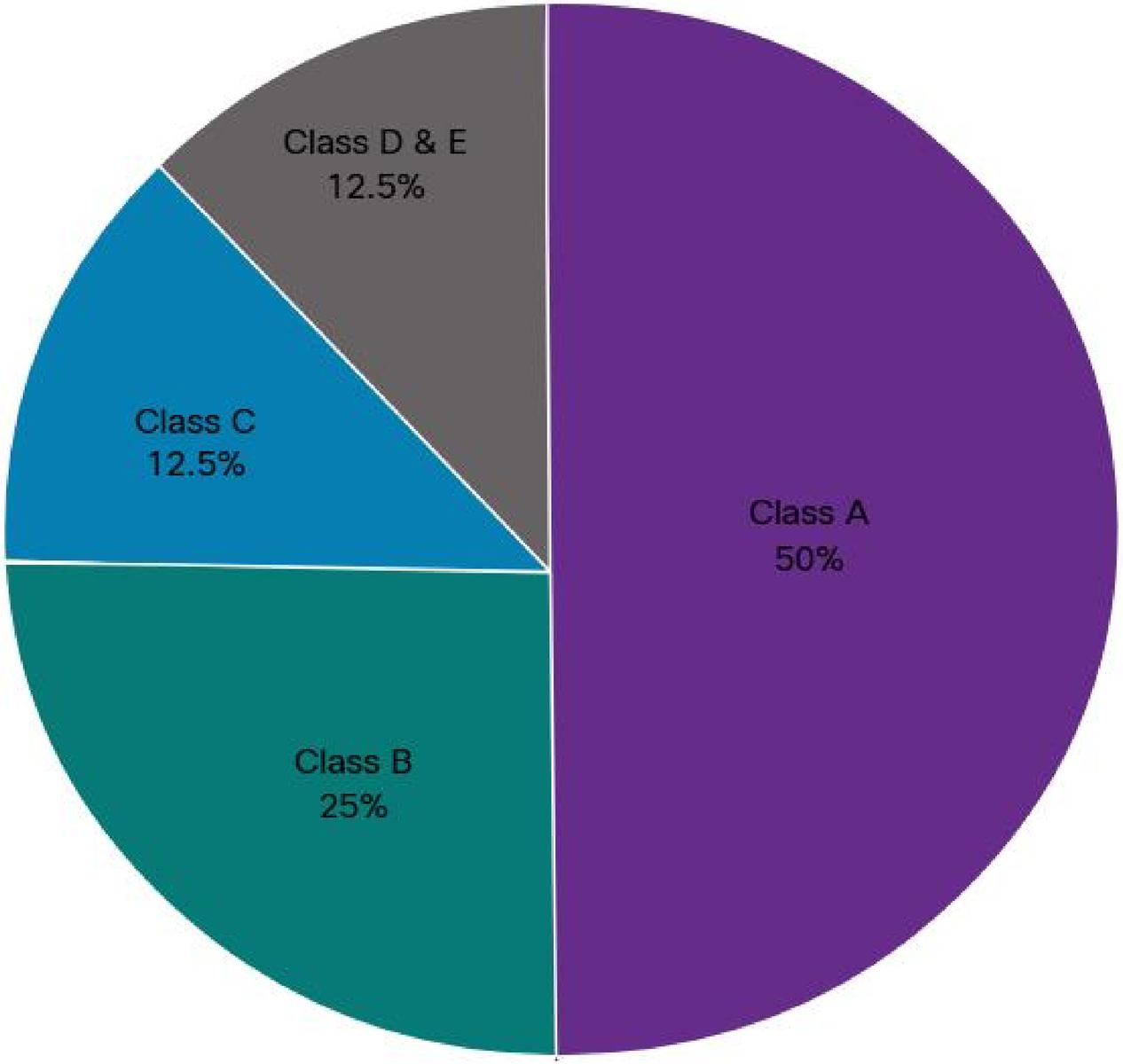
RFC 790 (1981) allocated IPv4 addresses in classes

- Class A (0.0.0.0/8 to 127.0.0.0/8)
- Class B (128.0.0.0 /16 – 191.255.0.0 /16)
- Class C (192.0.0.0 /24 – 223.255.255.0 /24)
- Class D (224.0.0.0 to 239.0.0.0)
- Class E (240.0.0.0 – 255.0.0.0)

- Classful addressing wasted many IPv4 addresses.

Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

Legacy Classful Addressing



Class A

Total Networks: 128
Total Hosts/Net: 16,777,214

Class B

Total Networks: 16,384
Total Hosts/Net: 65,534

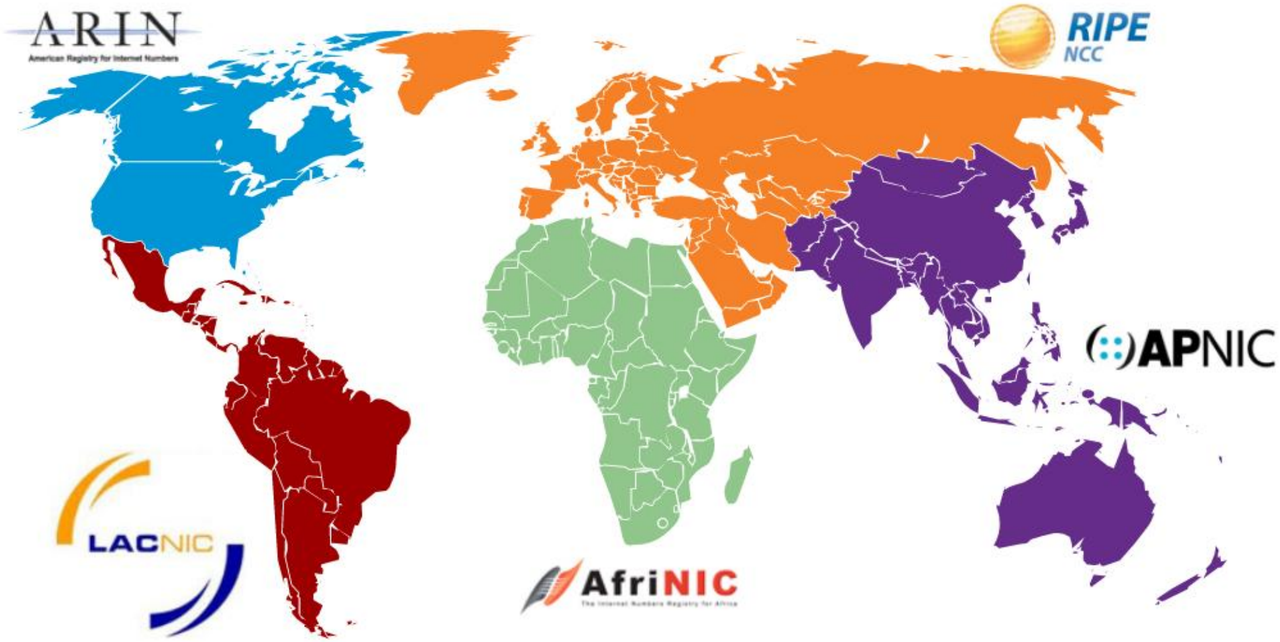
Class C

Total Networks: 2,097,152
Total Hosts/Net: 254

Assignment of IP Addresses

- The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).

- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.

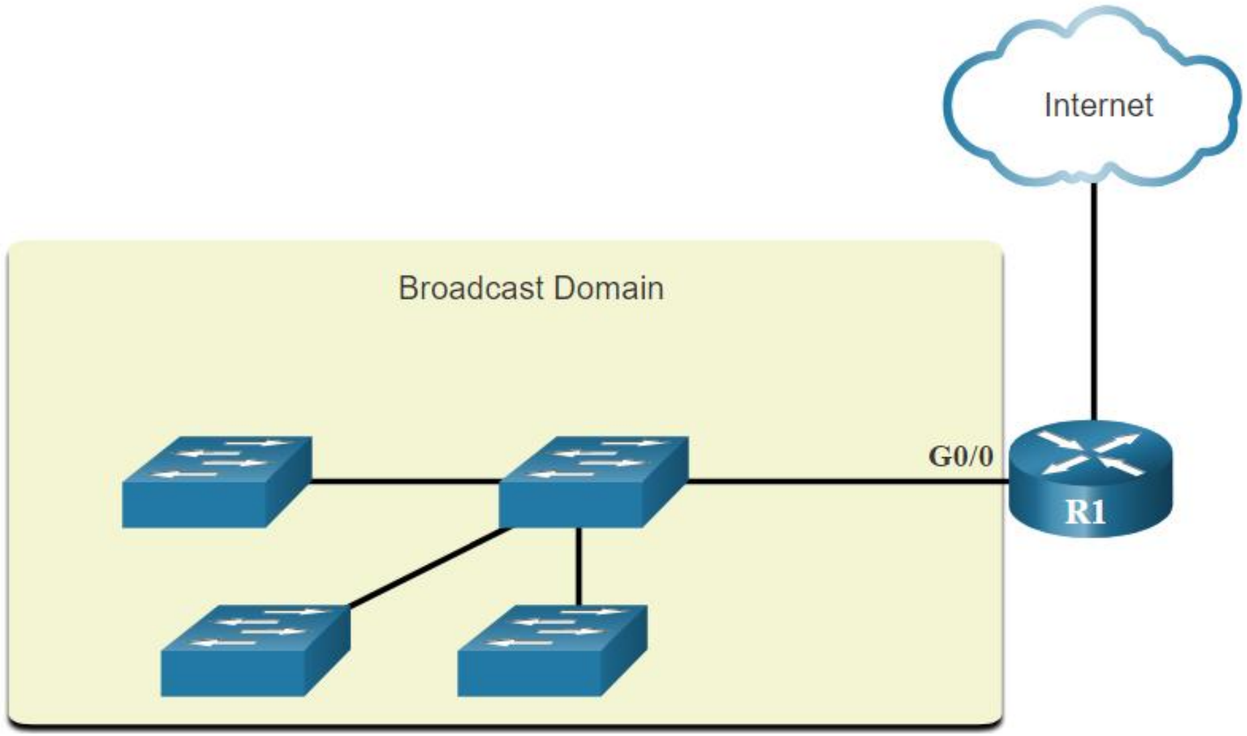


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Network Segmentation

Broadcast Domains and Segmentation

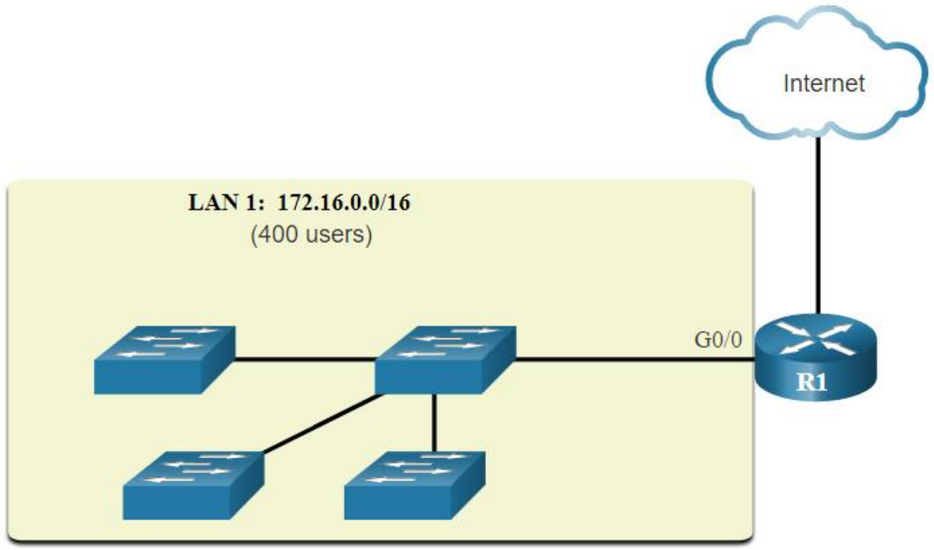
- Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
- Switches propagate broadcasts out all interfaces except the interface on which it was received.



- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.

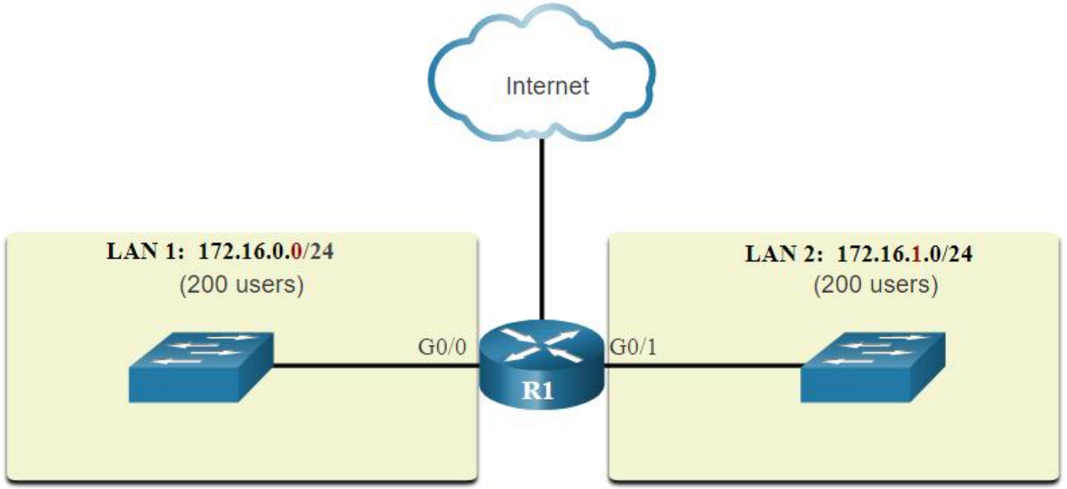
Problems with Large Broadcast Domains

- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.



- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.

- Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.
- Broadcasts are only propagated within the smaller broadcast domains.

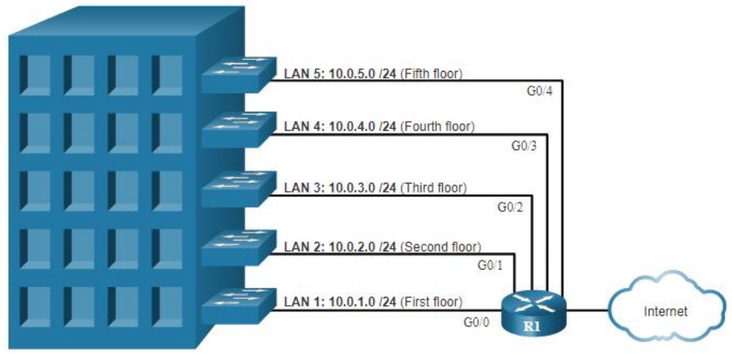


Reasons for Segmenting Networks

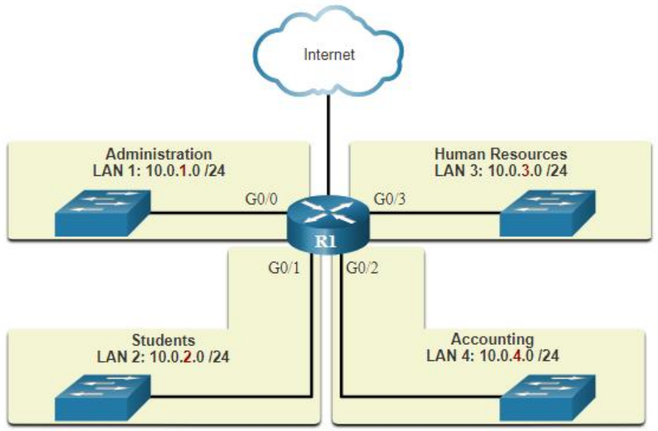
- Subnetting reduces overall network traffic and improves network performance.
- It can be used to implement security policies between subnets.
- Subnetting reduces the number of devices affected by abnormal broadcast traffic.

- Subnets are used for a variety of reasons including by:

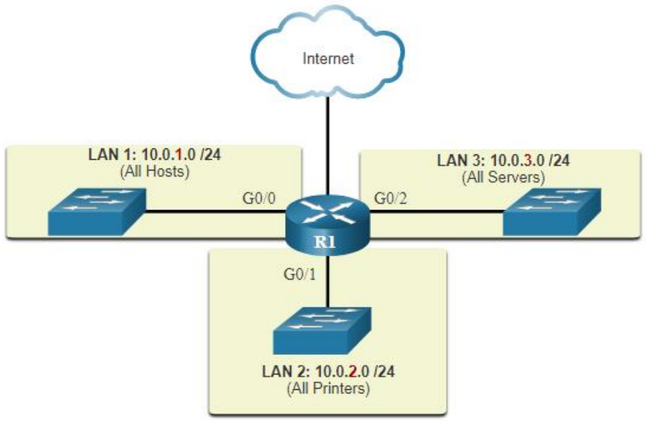
Location



Group or Function



Device Type



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Subnet an IPv4 Network

Subnet on an Octet Boundary

- Networks are most easily subnetted at the octet boundary of /8, /16, and /24.

- Notice that using longer prefix lengths decreases the number of hosts per subnet.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	**255**.0.0.0	**nnnnnnnn**.hhhhhhhh.hhhhhhhh.hhhhhhhh **11111111**.00000000.00000000.00000000	16,777,214
/16	**255.255**.0.0	**nnnnnnnn.nnnnnnnn**.hhhhhhhh.hhhhhhhh **11111111.11111111**.00000000.00000000	65,534
/24	**255.255.255**.0	**nnnnnnnn.nnnnnnnn.nnnnnnnn**.hhhhhhhh **11111111.11111111.11111111**.00000000	254

Subnet on an Octet Boundary (Cont.)

- In the first table 10.0.0.0/8 is subnetted using /16 and in the second table, a /24 mask.

Subnet Address (256 Possible Subnets)	**Host Range** (65,534 possible hosts per subnet)	**Broadcast**
10.0.0.0**/16**	**10.0**.0.1 - **10.0**.255.254	**10.0**.255.255
10.1.0.0**/16**	**10.1**.0.1 - **10.1**.255.254	**10.1**.255.255
10.2.0.0**/16**	**10.2**.0.1 - **10.2**.255.254	**10.2**.255.255
10.3.0.0**/16**	**10.3**.0.1 - **10.3**.255.254	**10.3**.255.255
10.4.0.0**/16**	**10.4**.0.1 - **10.4**.255.254	**10.4**.255.255
10.5.0.0**/16**	**10.5**.0.1 - **10.5**.255.254	**10.5**.255.255
10.6.0.0**/16**	**10.6**.0.1 - **10.6**.255.254	**10.6**.255.255
10.7.0.0**/16**	**10.7**.0.1 - **10.7**.255.254	**10.7**.255.255
…	…	…
10.255.0.0**/16**	**10.255**.0.1 - **10.255**.255.254	**10.255**.255.255

Subnet Address (65,536 Possible Subnets)	**Host Range** (254 possible hosts per subnet)	**Broadcast**
10.0.0.0**/24**	**10.0.0**.1 - **10.0.0**.254	**10.0.0**.255
10.0.1.0**/24**	**10.0.1**.1 - **10.0.1**.254	**10.0.1**.255
10.0.2.0**/24**	**10.0.2**.1 - **10.0.2**.254	**10.0.2**.255
…	…	…
10.0.255.0**/24**	**10.0.255**.1 - **10.0.255**.254	**10.0.255**.255
10.1.0.0**/24**	**10.1.0**.1 - **10.1.0**.254	**10.1.0**.255
10.1.1.0**/24**	**10.1.1**.1 - **10.1.1**.254	**10.1.1**.255
10.1.2.0**/24**	**10.1.2**.1 - **10.1.2**.254	**10.1.2**.255
…	…	…
10.100.0.0**/24**	**10.100.0**.1 - **10.100.0**.254	**10.100.0**.255
…	…	…
10.255.255.0**/24**	**10.255.255**.1 - **10.2255.255**.254	**10.255.255**.255

Subnet within an Octet Boundary

- Refer to the table to see six ways to subnet a /24 network.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.**n**hhhhhhh 11111111.11111111.11111111.**1**0000000	**2**	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nn**hhhhhh 11111111.11111111.11111111.**11**000000	**4**	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnn**hhhhh 11111111.11111111.11111111.**111**00000	**8**	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnn**hhhh 11111111.11111111.11111111.**1111**0000	**16**	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnn**hhh 11111111.11111111.11111111.**11111**000	**32**	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnnn**hh 11111111.11111111.11111111.**111111**00	**64**	2

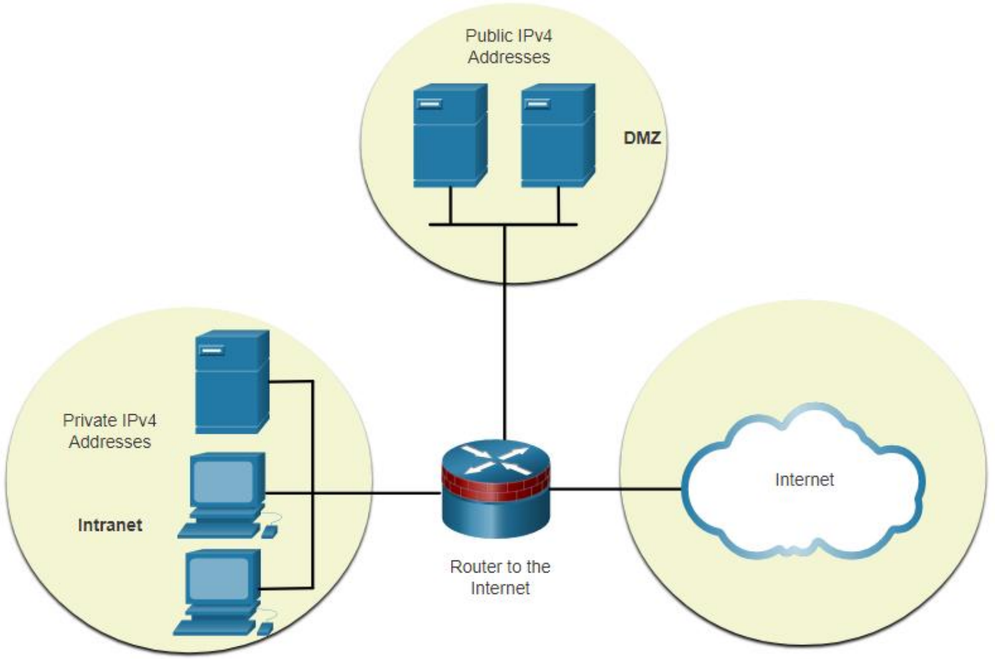
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Subnet to Meet Requirements

Subnet Private versus Public IPv4 Address Space

Enterprise networks will have an:

- Intranet - A company's internal network typically using private IPv4 addresses.
- DMZ – A companies internet facing servers. Devices in the DMZ use public IPv4 addresses.
- A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.
- The DMZ devices would have to be configured with public IP addresses.



Subnet to Meet Requirements

Minimize Unused Host IPv4 Addresses and Maximize Subnets

There are two considerations when planning subnets:

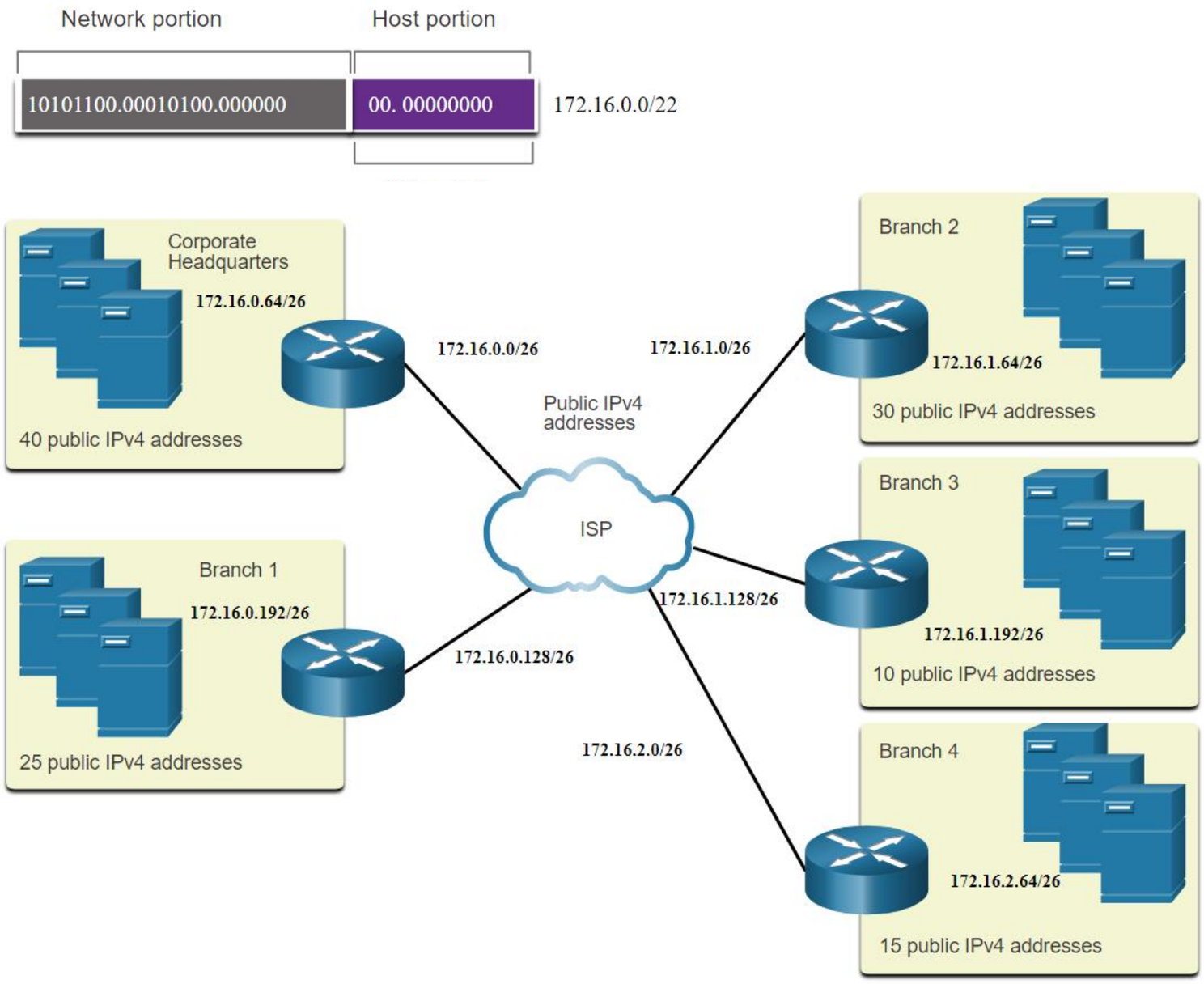
- The number of host addresses required for each network
- The number of individual subnets needed

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.**n**hhhhhhh 11111111.11111111.11111111.**1**0000000	**2**	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nn**hhhhhh 11111111.11111111.11111111.**11**000000	**4**	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnn**hhhhh 11111111.11111111.11111111.**111**00000	**8**	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnn**hhhh 11111111.11111111.11111111.**1111**0000	**16**	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnn**hhh 11111111.11111111.11111111.**11111**000	**32**	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnnn**hh 11111111.11111111.11111111.**111111**00	**64**	2

Subnet to Meet Requirements

Example: Efficient IPv4 Subnetting

- In this example, corporate headquarters has been allocated a public network address of 172.16.0.0/22 (10 host bits) by its ISP providing 1,022 host addresses.
- There are five sites and therefore five internet connections which means the organization requires 10 subnets with the largest subnet requires 40 addresses.
- It allocated 10 subnets with a /26 (i.e., 255.255.255.192) subnet mask.



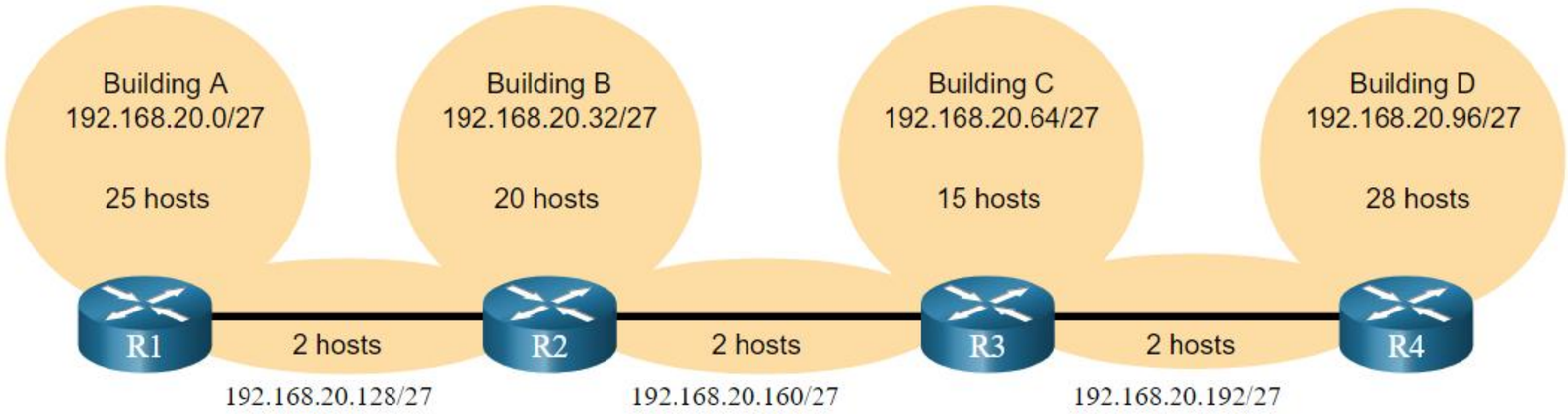
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VLSM

IPv4 Address Conservation

Given the topology, 7 subnets are required (i.e, four LANs and three WAN links) and the largest number of host is in Building D with 28 hosts.

- A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology.



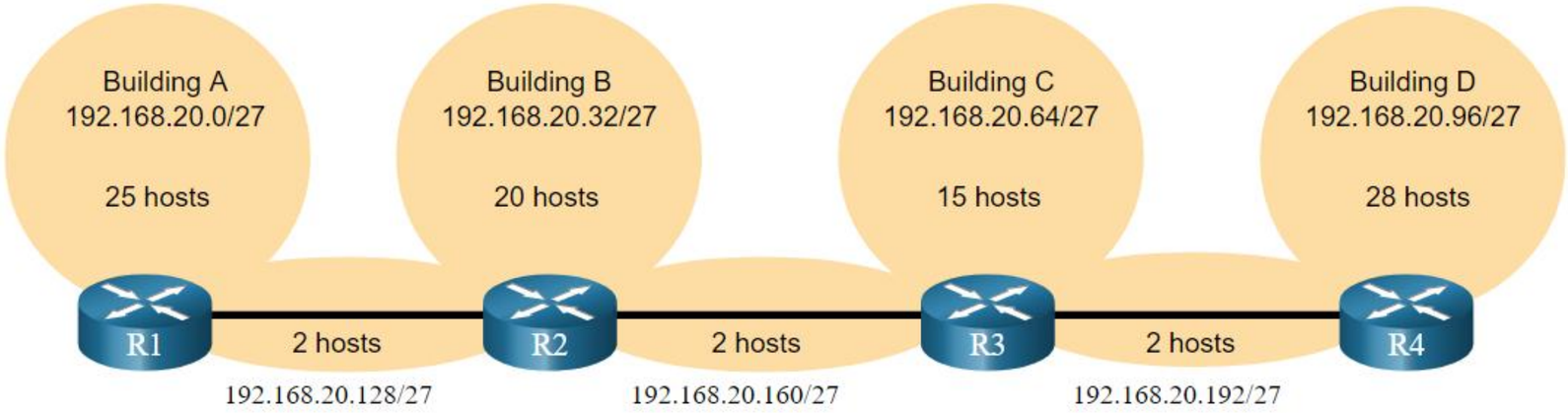
IPv4 Address Conservation (Cont.)

However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses.

Host portion
2^5 – 2 = 30 host IP addresses per subnet

30 – 2 = 28
Each WAN subnet wastes 28 addresses

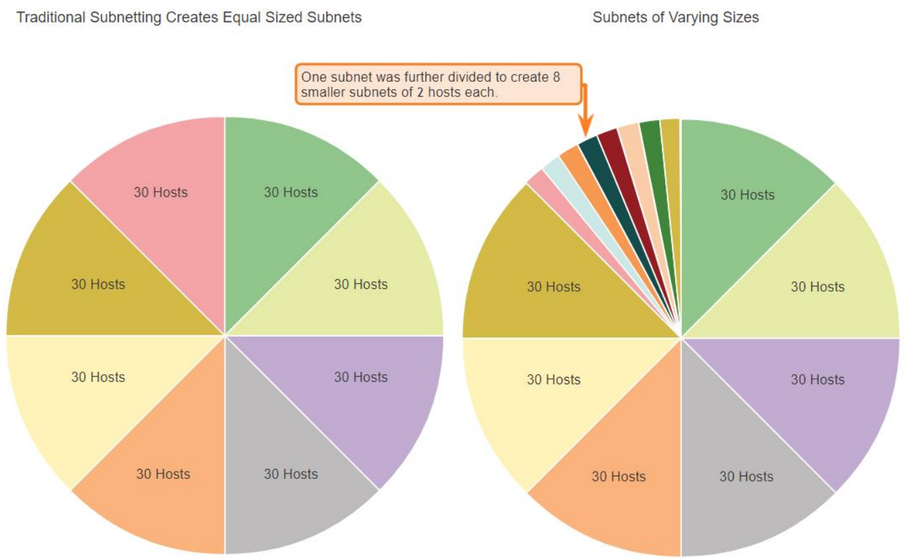
28 x 3 = 84
84 addresses are unused



- Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
- VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet.

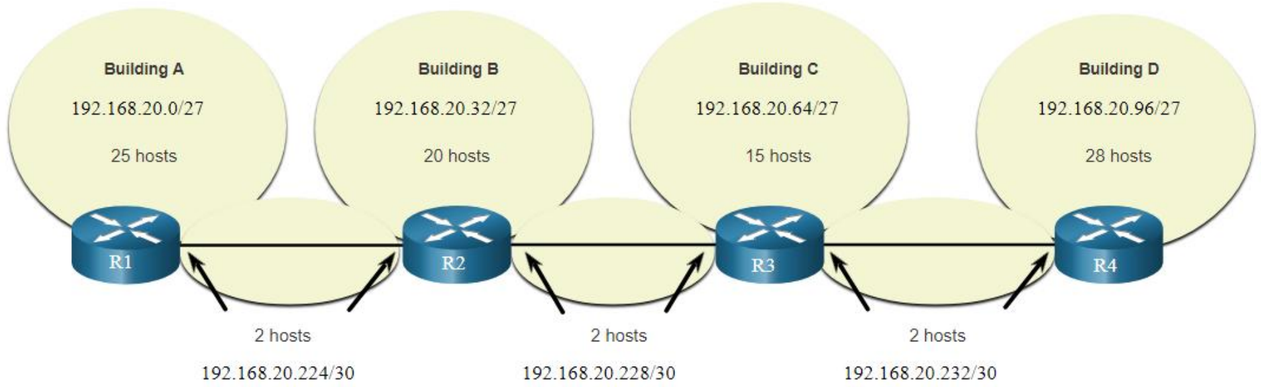
VLSM

- The left side displays the traditional subnetting scheme (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to subnet a subnet and divided the last subnet into eight /30 subnets.



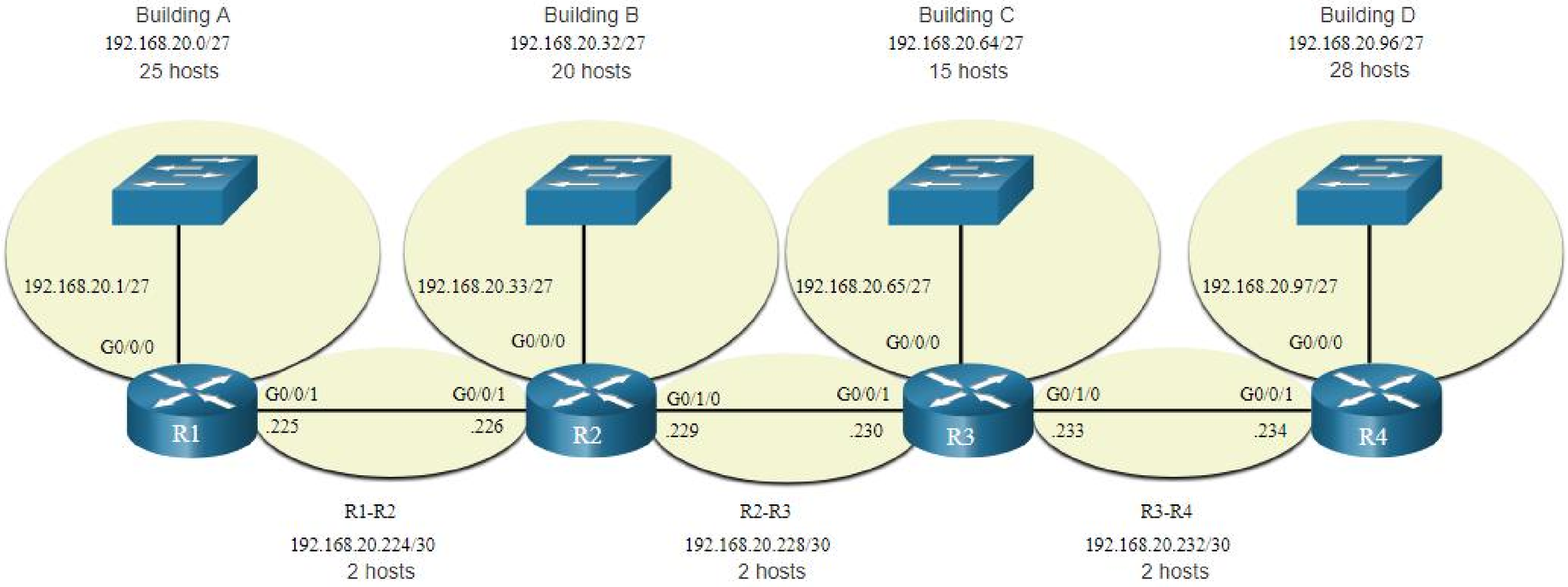
- When using VLSM, always begin by satisfying the host requirements of the largest subnet and continue subnetting until the host requirements of the smallest subnet are satisfied.

- The resulting topology with VLSM applied.



VLSM Topology Address Assignment

- Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram.



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Structured Design

IPv4 Network Address Planning

IP network planning is crucial to develop a scalable solution to an enterprise network.

- To develop an IPv4 network wide addressing scheme, you need to know how many subnets are needed, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses, and which use public, and many other determining factors.

Examine the needs of an organization’s network usage and how the subnets will be structured.

- Perform a network requirement study by looking at the entire network to determining how each area will be segmented.
- Determine how many subnets are needed and how many hosts per subnet.
- Determine DHCP address pools and Layer 2 VLAN pools.

Device Address Assignment

Within a network, there are different types of devices that require addresses:

- **End user clients** – Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.
- **Servers and peripherals** – These should have a predictable static IP address.
- **Servers that are accessible from the internet** – Servers must have a public IPv4 address, most often accessed using NAT.
- **Intermediary devices** – Devices are assigned addresses for network management, monitoring, and security.
- **Gateway** – Routers and firewall devices are gateway for the hosts in that network.

When developing an IP addressing scheme, it is generally recommended that you have a set pattern of how addresses are allocated to each type of device.

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THANK YOU!